HSC Math Extension 1: Polynomials Mastery

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The theory of equations is one of the most beautiful and important branches of mathematics. Every polynomial equation holds a secret waiting to be factored out.

Adapted from classical algebra

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1 Introduction

1.1 Project Overview

This booklet is a growing collection of polynomial problems for NSW HSC Mathematics Extension 1 students. It covers the full polynomial strand of the Extension 1 syllabus, including the language and notation of polynomials, graphs of polynomial functions, division algorithms, the remainder and factor theorems, consequences of the factor theorem, sums and products of zeroes (roots), and multiple zeroes. The problems range from straightforward skill-drill questions to multi-step challenge problems suitable for exam preparation.

1.2 Target Audience

This resource is written for students who want focused, exam-oriented practice with polynomials in a clear, classroom-friendly format. Solutions are intentionally concise but each one highlights the key algebraic idea behind the answer. Students who work systematically through Parts 1 and 2 should gain both fluency and deeper conceptual understanding of the polynomial topics tested in Extension 1.

1.3 How to Use This Booklet

- Read the core formulas and theory summaries in this introduction first.
- Attempt each problem before reading the solution.
- Pay attention to the *type* of question: is it a factoring question, a root-extraction question, or a proof using identities?
- Look for structure: symmetry among roots, repeated factors, or relationships that allow the factor theorem to be applied efficiently.
- Use the remarks and takeaways to build pattern recognition for future HSC questions.

1.4 Extension 1 Knowledge Needed

Most problems in this booklet can be solved using standard HSC Mathematics Extension 1 knowledge. The key background material is:

- **Polynomial notation:** understanding degree, leading coefficient, constant term, and the standard form
$$P(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0, \quad a_n \neq 0.$$
- **Equality of polynomials:** two polynomials are equal if and only if all corresponding coefficients are equal.
- **Division algorithm:** for polynomials $P(x)$ and $D(x)$ with $\deg D \leq \deg P$, there exist unique $Q(x)$ and $R(x)$ with $\deg R < \deg D$ such that $P(x) = D(x)Q(x) + R(x)$.
- **Remainder theorem:** when $P(x)$ is divided by $(x - a)$, the remainder equals $P(a)$.
- **Factor theorem:** $(x - a)$ is a factor of $P(x)$ if and only if $P(a) = 0$.
- **Vieta's formulas (sums and products of roots):** for $P(x) = x^n + p_{n-1}x^{n-1} + \cdots + p_0$, the elementary symmetric functions of the roots $\alpha_1, \ldots, \alpha_n$ are expressible in terms of the coefficients.

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- **Multiple zeroes:** a is a zero of multiplicity m when $(x-a)^m \mid P(x)$ but $(x-a)^{m+1} \nmid P(x)$; equivalently, $P(a) = P'(a) = \cdots = P^{(m-1)}(a) = 0$ and $P^{(m)}(a) \neq 0$.
- **Graph behaviour near roots:** the curve crosses the x-axis at simple roots and touches (but does not cross) at even-multiplicity roots.
- **Long division and synthetic division:** systematic methods for finding quotients and remainders.
- **Sketching polynomial graphs:** leading-term behaviour, y-intercept, x-intercepts, and turning points.

1.5 Core Formulas and Ideas

Definition of a Polynomial. A **polynomial** in x of degree n is an expression of the form

$$P(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0,$$

where $a_n, a_{n-1}, \ldots, a_0$ are real (or complex) constants and $a_n \neq 0$. The highest power n is the **degree** of P, and a_n is its **leading coefficient**.

Polynomial Equality. Two polynomials are equal if and only if they have the same degree and all corresponding coefficients are equal:

$$a_n x^n + \cdots + a_0 = b_n x^n + \cdots + b_0 \iff a_k = b_k \text{ for all } k.$$

Division Algorithm for Polynomials. Given polynomials $P(x)$ and $D(x)$ with $D(x) \neq 0$ and $\deg D \leq \deg P$, there exist unique polynomials $Q(x)$ (the *quotient*) and $R(x)$ (the *remainder*) such that

$$P(x) = D(x)\,Q(x) + R(x), \qquad \deg R < \deg D \text{ (or } R = 0).$$

Remainder Theorem. When the polynomial $P(x)$ is divided by the linear factor $(x-a)$, the remainder is

$$R = P(a).$$

Factor Theorem. $(x-a)$ is a factor of $P(x)$ if and only if

$$P(a) = 0.$$

More generally, $(ax-b)$ is a factor of $P(x)$ if and only if $P\left(\frac{b}{a}\right) = 0$.

1.6 Vieta's Formulas (Sums and Products of Roots)

Let $P(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_0$ have roots $\alpha_1, \alpha_2, \ldots, \alpha_n$. Then:

1. **Sum of roots:**
$$\alpha_1 + \alpha_2 + \cdots + \alpha_n = -\frac{a_{n-1}}{a_n}.$$
2. **Sum of products of pairs:**
$$\sum_{i<j} \alpha_i \alpha_j = \frac{a_{n-2}}{a_n}.$$
3. **Product of all roots:**
$$\alpha_1\,\alpha_2 \cdots \alpha_n = (-1)^n \frac{a_0}{a_n}.$$

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Quadratic case. If $P(x) = x^2 + bx + c$ has roots α, β:

$$\alpha + \beta = -b, \qquad \alpha\beta = c.$$

Cubic case. If $P(x) = x^3 + px^2 + qx + r$ has roots α, β, γ:

$$\alpha + \beta + \gamma = -p, \qquad \alpha\beta + \beta\gamma + \gamma\alpha = q, \qquad \alpha\beta\gamma = -r.$$

1.7 Multiple Zeroes and Graph Behaviour

A number a is a zero of **multiplicity** m of the polynomial $P(x)$ if

$$(x-a)^m \mid P(x) \quad \text{but} \quad (x-a)^{m+1} \nmid P(x).$$

An equivalent characterisation using derivatives:

$$P(a) = P'(a) = \cdots = P^{(m-1)}(a) = 0, \quad P^{(m)}(a) \neq 0.$$

Graph shape at a zero:

- **Simple zero** ($m = 1$): the graph *crosses* the x-axis.
- **Double zero** ($m = 2$): the graph *touches* the x-axis and turns back (like $y = x^2$ at the origin).
- **Triple zero** ($m = 3$): the graph *flattens* and crosses the x-axis (like $y = x^3$ at the origin).

1.8 Sketching Polynomial Graphs

Key features to determine when sketching $y = P(x)$:

1. **Degree and leading coefficient:** determines end behaviour.
   - Even degree, positive leading coefficient: both ends go to $+\infty$.
   - Even degree, negative leading coefficient: both ends go to $-\infty$.
   - Odd degree, positive leading coefficient: left end $\to -\infty$, right end $\to +\infty$.
   - Odd degree, negative leading coefficient: left end $\to +\infty$, right end $\to -\infty$.
2. **y-intercept:** $P(0) = a_0$.
3. **x-intercepts:** the real roots of $P(x) = 0$, taking multiplicities into account.
4. **Turning points:** at most $n - 1$ turning points for a degree-n polynomial.

1.9 Consequences of the Factor Theorem

- If a polynomial $P(x)$ of degree n has n distinct real roots $\alpha_1, \ldots, \alpha_n$, then

$$P(x) = a_n(x - \alpha_1)(x - \alpha_2)\cdots(x - \alpha_n).$$

- A polynomial of degree n has *at most* n roots (counted with multiplicity) unless it is the zero polynomial.
- If two polynomials of degree $\leq n$ agree at $n + 1$ distinct points, they must be identical.
- **Finding factors:** to factor $P(x)$, test rational candidates $\pm\frac{p}{q}$ where $p \mid a_0$ and $q \mid a_n$ (rational root theorem), then use polynomial division to find the remaining factors.

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1.10 Division Techniques

Long division. Systematically divide the leading term of the dividend by the leading term of the divisor, multiply, subtract, and repeat until the degree of the remainder is less than the degree of the divisor.

Synthetic division (for linear divisors). A streamlined numerical method for dividing $P(x)$ by $(x - a)$; it records only the coefficients and uses the value a as the divisor.

Example. To divide $P(x) = 2x^3 - 3x^2 + x - 5$ by $(x - 2)$ using synthetic division:

$$\begin{array}{c|cccc} 2 & 2 & -3 & 1 & -5 \\ \hline & & 4 & 2 & 6 \\ \hline & 2 & 1 & 3 & 1 \end{array}$$

So $P(x) = (x - 2)(2x^2 + x + 3) + 1$, and the remainder is $P(2) = 1$. ✓

1.11 Problem-Solving Habits for This Booklet

- Always identify the *degree* and *leading coefficient* first.
- When using the factor theorem, try $x = \pm 1, \pm 2, \ldots$ before turning to the rational root theorem.
- For root-sum/product problems, write out Vieta's formulas explicitly before attempting algebra.
- For multiple-zero problems, consider taking derivatives to set up a system of equations.
- For graph sketching, mark each intercept with its multiplicity and check end behaviour.
- In proof questions, compare coefficients or use the identity principle for polynomials.

Remark 1.1

The most common polynomial mistake is forgetting to account for multiplicity: a polynomial of degree n may have *fewer* than n distinct real roots, but it always has *exactly* n roots when complex roots and multiplicities are counted together (Fundamental Theorem of Algebra).

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2 Part 1: Worked Problems

2.1 Basic

2.2 Medium

2.3 Advanced

3 Part 2: Practice and Challenge Bank

3.1 Warm-Up Drills

3.2 Stretch Problems

3.3 Challenge Corner

4 Conclusion

Polynomials reward systematic thinking as much as algebraic fluency. Many HSC Extension 1 questions become far more tractable once you identify the correct strategy: applying the factor theorem, writing out Vieta's formulas, or reading the graph multiplicity from first principles. As this booklet grows, it will serve as a compact reference for the most important polynomial patterns students encounter in Extension 1 examinations and beyond.

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